

Contrastive Learning Methods

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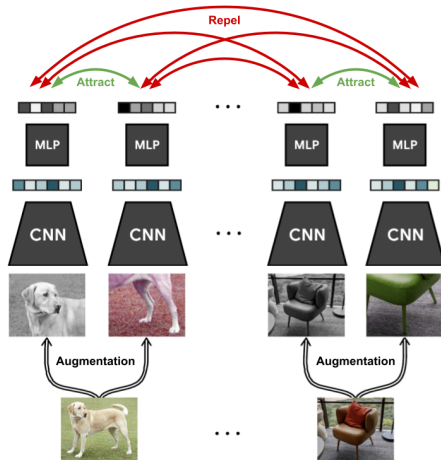
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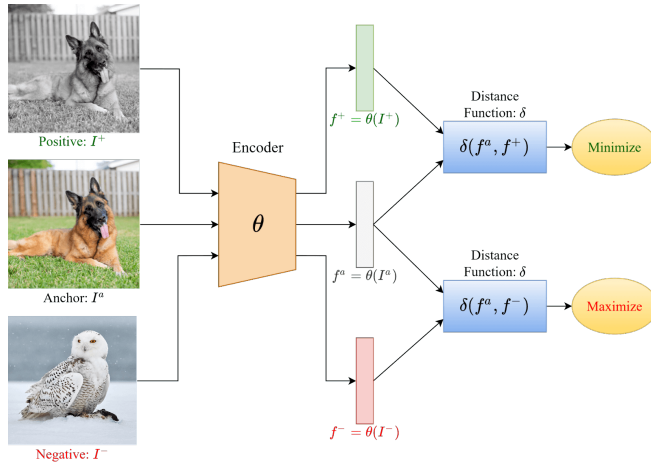
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Source: UvA Deep Learning Tutorials (P. Lippe), SimCLR tutorial; figure credit Chen et al. (SimCLR).

1. Simple Framework for Contrastive Learning (SimCLR)
2. Bootstrap Your Own Latent (BYOL)
3. Distillation with No Labels (DINO)
4. DINOv2: Improved DINO
5. Image BERT (iBOT)
6. Weakly Supervised Learning (WSL)
7. Summary



[Source:

<https://framerusercontent.com/images/BTYwexvG8pobxihJcbBy0Vp4aE.png>]

Where We Are

- ▶ Pretext tasks enable self-supervised learning by creating artificial labels.
- ▶ Instance discrimination treats each image as its own class, encouraging unique representations.
- ▶ MoCo framework uses a dynamic dictionary with a queue and momentum encoder for contrastive learning.
- ▶ **Limitations of MoCo:**
 - Need for large memory banks
 - Complexity of momentum encoders
 - Sensitivity to negative sampling

Why Evolve?

- ▶ Wouldn't it be great if we could get rid of all that extra memory and complicated tricks?
- ▶ Can we squeeze even more out of the positive pairs we already have?
- ▶ What if our models could focus on the big picture, not just tiny pixel details?

By the end of this lecture, you will be able to:

- ▶ Trace how contrastive learning evolved from MoCo → SimCLR → BYOL → DINO → iBOT.
- ▶ Identify the design trade-offs: memory vs. compute, and negative sampling vs. representational collapse.
- ▶ Explain how batch size and temperature (τ) shape the learned embeddings.
- ▶ Explain why the projection head is dropped when moving to downstream tasks.
- ▶ Analyze how BYOL avoids representational collapse without negative pairs.
- ▶ Illustrate why centering and sharpening the teacher's outputs prevent collapse in DINO.

Contrastive Learning: **Simple Framework for Contrastive Learning (SimCLR)**

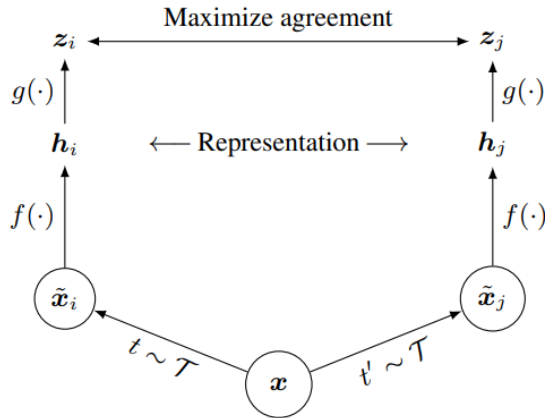
A Simple Framework for Contrastive Learning of Visual Representations

Ting Chen¹ Simon Kornblith¹ Mohammad Norouzi¹ Geoffrey Hinton¹

What is SimCLR?

- ▶ A simple and powerful method for learning image features without labels.
- ▶ Uses contrastive learning:
 - Pulls similar images (augmentations of the same image) closer together.
 - Pushes different images (from different sources) further apart.

SimCLR: Simple Framework for Contrastive Learning (cont.)



Source: Chen, Kornblith, Norouzi & Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," ICML 2020, Fig. 2.

Why was SimCLR created?

- ▶ Earlier contrastive learning methods needed extra tricks:
 - Memory banks to store negative examples.
 - Special momentum encoders.
- ▶ SimCLR avoids these by:
 - Using very large batch sizes.
 - Always having plenty of negative examples in each batch.
- ▶ This makes SimCLR simpler and easier to train.

How does SimCLR work?

► Augmentation pipeline:

- Each image is randomly cropped.
- Color is changed.
- Sometimes blurred.
- This creates two different views of the same image.

► Encoder + Projection Head:

- Both views go through a neural network encoder ($f(\cdot)$).
- Then, they pass through a small MLP called the projection head ($g(\cdot)$).
- This gives the final representations.

► Contrastive Loss (NT-Xent):

- The model pulls together representations of two views of the same image.
- It pushes apart representations of different images.
- The loss function is:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

- $\text{sim}(z_i, z_j)$ is the similarity between two representations.
- τ is a temperature parameter.

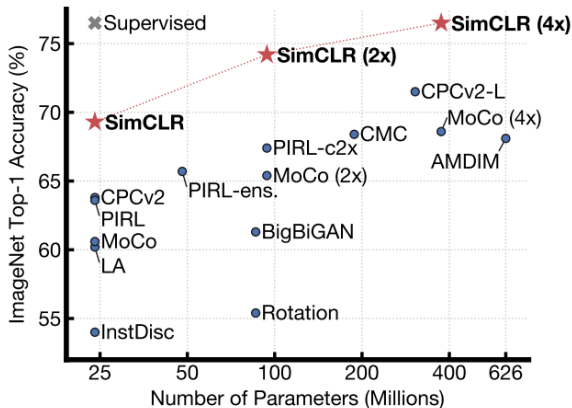
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return encoder_network f

```

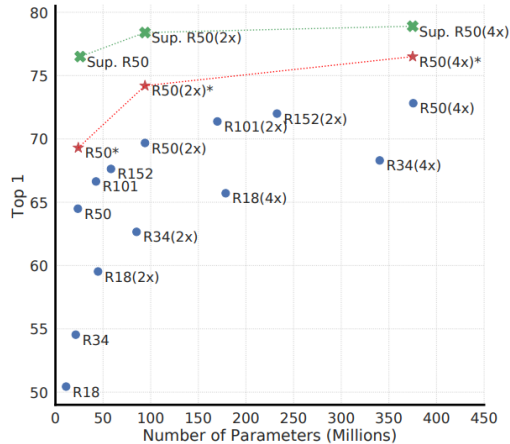
16

SimCLR: Simple Framework for Contrastive Learning (cont.)



Source: Chen, Kornblith, Norouzi & Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," ICML 2020, Fig. 1.

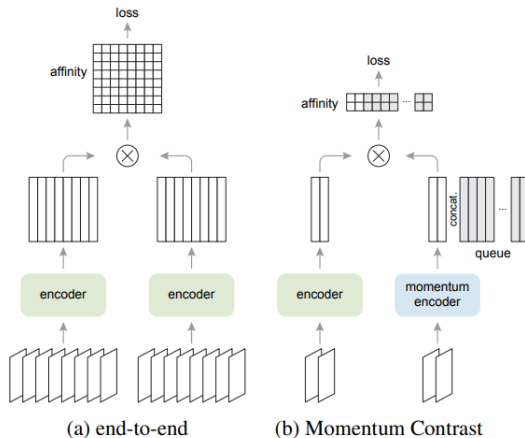
SimCLR: Simple Framework for Contrastive Learning (cont.)



Source: Chen, Kornblith, Norouzi & Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," ICML 2020, Fig. 7.

What did we learn from SimCLR?

- ▶ Using strong data augmentations is really important for learning good features.
- ▶ Adding a nonlinear projection head (the MLP) helps the model learn better representations.
- ▶ SimCLR needs large batch sizes, which means it requires a lot of computing power.



Source: Chen, Fan, Girshick & He, “Improved Baselines with Momentum Contrastive Learning,”
arXiv:2003.04297, 2020, Fig. 1.

case	unsup. pre-train				ImageNet acc.	VOC detection		
	MLP	aug+	cos	epochs		AP ₅₀	AP	AP ₇₅
supervised					76.5	81.3	53.5	58.8
MoCo v1				200	60.6	81.5	55.9	62.6
(a)	✓			200	66.2	82.0	56.4	62.6
(b)		✓		200	63.4	82.2	56.8	63.2
(c)	✓	✓		200	67.3	82.5	57.2	63.9
(d)	✓	✓	✓	200	67.5	82.4	57.0	63.6
(e)	✓	✓	✓	800	71.1	82.5	57.4	64.0

Source: Chen, Fan, Girshick & He, “Improved Baselines with Momentum Contrastive Learning,” arXiv:2003.04297, 2020, Table 1.

case	unsup. pre-train					ImageNet acc.
	MLP	aug+	cos	epochs	batch	
MoCo v1 [6]				200	256	60.6
SimCLR [2]	✓	✓	✓	200	256	61.9
SimCLR [2]	✓	✓	✓	200	8192	66.6
MoCo v2	✓	✓	✓	200	256	67.5
<i>results of longer unsupervised training follow:</i>						
SimCLR [2]	✓	✓	✓	1000	4096	69.3
MoCo v2	✓	✓	✓	800	256	71.1

Source: Chen, Fan, Girshick & He, “Improved Baselines with Momentum Contrastive Learning,”
arXiv:2003.04297, 2020, Table 2.

Contrastive Learning: **Bootstrap Your Own Latent (BYOL)**

Bootstrap Your Own Latent A New Approach to Self-Supervised Learning

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Abstract

We introduce **Bootstrap Your Own Latent (BYOL)**, a new approach to self-supervised image representation learning. BYOL relies on two neural networks, referred to as *online* and *target* networks, that interact and learn from each other. From an augmented view of an image, we train the online network to predict the target network representation of the same image under a different augmented view. At the same time, we update the target network with a slow-moving average of the online network. While state-of-the-art methods intrinsically rely on negative pairs, BYOL achieves a new state of the art *without them*. BYOL reaches 74.3% top-1 classification accuracy on ImageNet using the standard linear evaluation protocol with a ResNet-50 architecture and 79.6% with a larger ResNet. We show that BYOL performs on par or better than the current state of the art on both transfer and semi-supervised benchmarks.

Why BYOL?

- ▶ Most contrastive methods use both positive and negative pairs.
- ▶ BYOL asks: Can we learn good features without negative pairs?
- ▶ BYOL shows it is possible to skip negatives and still learn useful representations.

BYOL (Bootstrap Your Own Latent) (cont.)

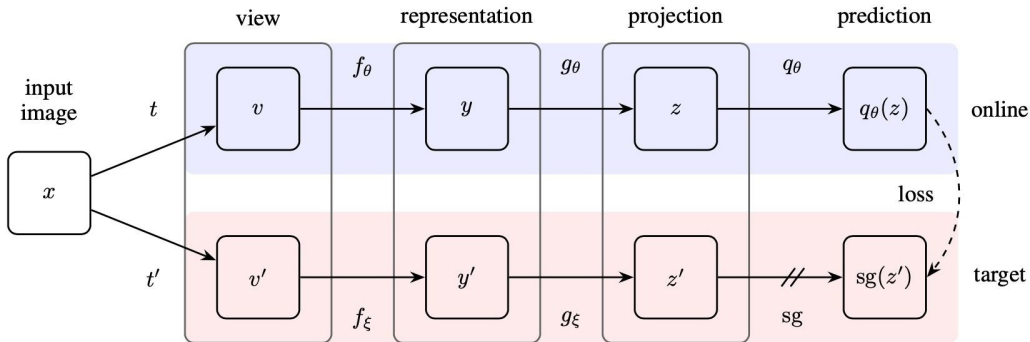


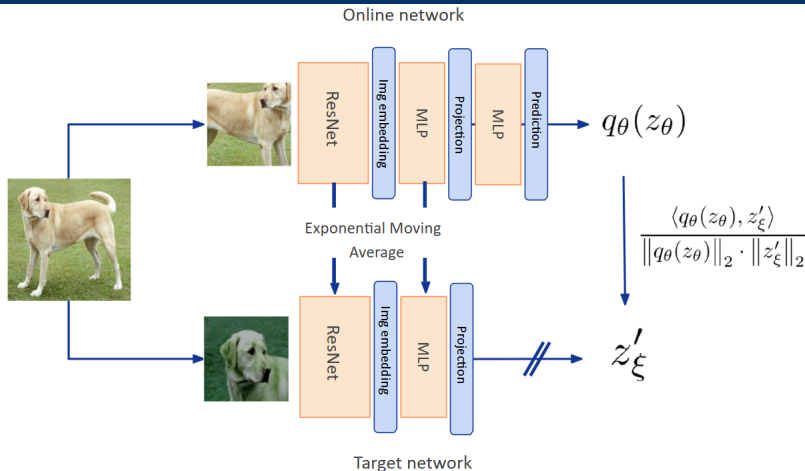
Figure 2: BYOL's architecture. BYOL minimizes a similarity loss between $q_\theta(z)$ and $sg(z')$, where θ are the trained weights, ξ are an exponential moving average of θ and sg means stop-gradient. At the end of training, everything but f_θ is discarded and y is used as the image representation.

Source: Grill et al., "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning," NeurIPS 2020, Fig. 2.

How does BYOL work?

- ▶ There are two networks: an **online network** and a **target network**. Both start out the same.
- ▶ Each network has an encoder (think of it as a feature extractor).
- ▶ The online network has an extra part called the **predictor head**.
- ▶ The target network's parameters are updated slowly using the online network's parameters (using something called an exponential moving average).
- ▶ The goal is simple: make the online network's prediction as close as possible to the target network's output, using mean squared error between their normalized outputs.

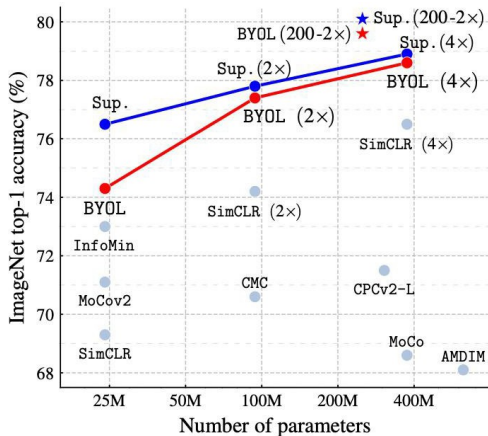
BYOL (Bootstrap Your Own Latent) (cont.)



Source: ResearchGate, "BYOL neural architecture"

(https://www.researchgate.net/figure/BYOL-neural-architecture_fig5_368988438).

BYOL (Bootstrap Your Own Latent) (cont.)



Source: Grill et al., "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning," NeurIPS 2020, Fig. 1.

Method	Top-1	Top-5
Local Agg.	60.2	-
PIRL [35]	63.6	-
CPC v2 [32]	63.8	85.3
CMC [11]	66.2	87.0
SimCLR [8]	69.3	89.0
MoCo v2 [37]	71.1	-
InfoMin Aug. [12]	73.0	91.1
BYOL (ours)	74.3	91.6

(a) ResNet-50 encoder.

Method	Architecture	Param.	Top-1	Top-5
SimCLR [8]	ResNet-50 (2→)	94M	74.2	92.0
CMC [11]	ResNet-50 (2→)	94M	70.6	89.7
BYOL (ours)	ResNet-50 (2→)	94M	77.4	93.6
CPC v2 [32]	ResNet-161	305M	71.5	90.1
MoCo [9]	ResNet-50 (4→)	375M	68.6	-
SimCLR [8]	ResNet-50 (4→)	375M	76.5	93.2
BYOL (ours)	ResNet-50 (4→)	375M	78.6	94.2
BYOL (ours)	ResNet-200 (2→)	250M	79.6	94.8

(b) Other ResNet encoder architectures.

Table 1: Top-1 and top-5 accuracies (in %) under linear evaluation on ImageNet.

Source: Grill et al., “Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning,” NeurIPS 2020, Table 1.

Method	Food101	CIFAR10	CIFAR100	Birdsnap	SUN397	Cars	Aircraft	VOC2007	DTD	Pets	Caltech-101	Flowers
<i>Linear evaluation:</i>												
BYOL (ours)	75.3	91.3	78.4	57.2	62.2	67.8	60.6	82.5	75.5	90.4	94.2	96.1
SimCLR (repro)	72.8	90.5	74.4	42.4	60.6	49.3	49.8	81.4	75.7	84.6	89.3	92.6
SimCLR [8]	68.4	90.6	71.6	37.4	58.8	50.3	50.3	80.5	74.5	83.6	90.3	91.2
Supervised-IN [8]	72.3	93.6	78.3	53.7	61.9	66.7	61.0	82.8	74.9	91.5	94.5	94.7
<i>Fine-tuned:</i>												
BYOL (ours)	88.5	97.8	86.1	76.3	63.7	91.6	88.1	85.4	76.2	91.7	93.8	97.0
SimCLR (repro)	87.5	97.4	85.3	75.0	63.9	91.4	87.6	84.5	75.4	89.4	91.7	96.6
SimCLR [8]	88.2	97.7	85.9	75.9	63.5	91.3	88.1	84.1	73.2	89.2	92.1	97.0
Supervised-IN [8]	88.3	97.5	86.4	75.8	64.3	92.1	86.0	85.0	74.6	92.1	93.3	97.6
Random init [8]	86.9	95.9	80.2	76.1	53.6	91.4	85.9	67.3	64.8	81.5	72.6	92.0

Table 3: Transfer learning results from ImageNet (IN) with the standard ResNet-50 architecture.

Source: Grill et al., “Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning,” NeurIPS 2020, Table 3.

What's special about BYOL?

- ▶ Even though there are no explicit negatives, the asymmetry between the online and target networks (the predictor plus the slowly updated target) prevents collapse.
- ▶ This means we don't need huge batch sizes or to search for negative samples.
- ▶ BYOL makes self-supervised learning simpler and more efficient!

Contrastive Learning: **Distillation with No Labels (DINO)**

Emerging Properties in Self-Supervised Vision Transformers

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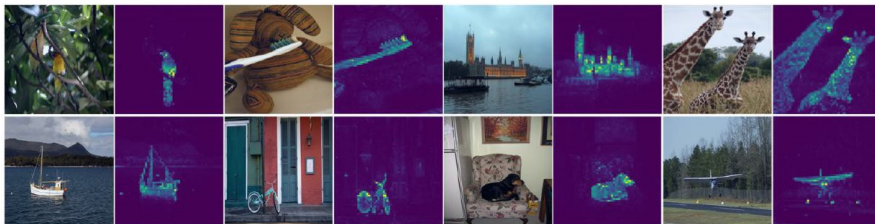


Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

Why DINO?

- ▶ Learns useful features from images without any labels.
- ▶ Helps the model understand both the big picture and small details.
- ▶ Shows that self-distillation alone, with no labels or negatives, produces strong features.

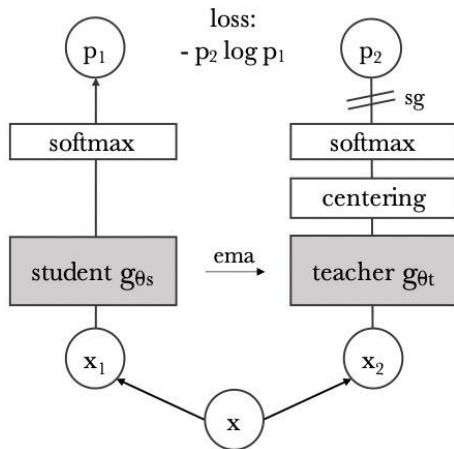


Figure 2: **Self-distillation with no labels.** We illustrate DINO in the case of one single pair of views (x_1, x_2) for simplicity. The model passes two different random transformations of an input image to the student and teacher networks. Both networks have the same architecture but different parameters. The output of the teacher network is centered with a mean computed over the batch. Each networks outputs a K dimensional feature that is normalized with a temperature softmax over the feature dimension. Their similarity is then measured with a cross-entropy loss. We apply a stop-gradient (sg) operator on the teacher to propagate gradients only through the student. The teacher parameters are updated with an exponential moving average (ema) of the student parameters.

Source: Caron et al., “Emerging Properties in Self-Supervised Vision Transformers,” ICCV 2021, Fig. 2.

How does DINO work?

► Student and Teacher Networks:

There are two networks with the same design. The teacher is just a slowly updated version of the student (using EMA).

► Multi-crop Strategy:

The model looks at the same image in different ways (two large views and several small crops) to learn features that work at different scales.

► No Negatives Needed:

Instead of comparing with negative samples, DINO uses a cross-entropy loss on soft similarity scores between the student and teacher outputs.

► Avoiding Collapse:

To make sure the model doesn't just output the same thing for every image, DINO centers and sharpens the teacher's outputs.

Algorithm 1 DINO PyTorch pseudocode w/o multi-crop.

```
# gs, gt: student and teacher networks
# C: center (K)
# tps, tpt: student and teacher temperatures
# l, m: network and center momentum rates
gt.params = gs.params
for x in loader: # load a minibatch x with n samples
    x1, x2 = augment(x), augment(x) # random views

    s1, s2 = gs(x1), gs(x2) # student output n-by-K
    t1, t2 = gt(x1), gt(x2) # teacher output n-by-K

    loss = H(t1, s2)/2 + H(t2, s1)/2
    loss.backward() # back-propagate

    # student, teacher and center updates
    update(gs) # SGD
    gt.params = l*gt.params + (1-l)*gs.params
    C = m*C + (1-m)*cat([t1, t2]).mean(dim=0)

def H(t, s):
    t = t.detach() # stop gradient
    s = softmax(s / tps, dim=1)
    t = softmax((t - C) / tpt, dim=1) # center + sharpen
    return - (t * log(s)).sum(dim=1).mean()
```

Source: Caron et al., “Emerging Properties in Self-Supervised Vision Transformers,” ICCV 2021, Algorithm 1.

Table 2: **Linear and k -NN classification on ImageNet.** We report top-1 accuracy for linear and k -NN evaluations on the validation set of ImageNet for different self-supervised methods. We focus on ResNet-50 and ViT-small architectures, but also report the best results obtained across architectures. * are run by us. We run the k -NN evaluation for models with official released weights. The throughput (im/s) is calculated on a NVIDIA V100 GPU with 128 samples per forward. Parameters (M) are of the feature extractor.

Method	Arch.	Param.	im/s	Linear	k -NN
Supervised	RN50	23	1237	79.3	79.3
SCLR [11]	RN50	23	1237	69.1	60.7
MoCov2 [13]	RN50	23	1237	71.1	61.9
InfoMin [54]	RN50	23	1237	73.0	65.3
BarlowT [66]	RN50	23	1237	73.2	66.0
OBoW [21]	RN50	23	1237	73.8	61.9
BYOL [23]	RN50	23	1237	74.4	64.8
DCv2 [9]	RN50	23	1237	75.2	67.1
SwAV [9]	RN50	23	1237	75.3	65.7
DINO	RN50	23	1237	75.3	67.5
Supervised	ViT-S	21	1007	79.8	79.8
BYOL* [23]	ViT-S	21	1007	71.4	66.6
MoCov2* [13]	ViT-S	21	1007	72.7	64.4
SwAV* [9]	ViT-S	21	1007	73.5	66.3
DINO	ViT-S	21	1007	77.0	74.5

Comparison across architectures

SCLR [11]	RN50w4	375	117	76.8	69.3
SwAV [9]	RN50w2	93	384	77.3	67.3
BYOL [23]	RN50w2	93	384	77.4	–
DINO	ViT-B/16	85	312	78.2	76.1
SwAV [9]	RN50w5	586	76	78.5	67.1
BYOL [23]	RN50w4	375	117	78.6	–
BYOL [23]	RN200w2	250	123	79.6	73.9
DINO	ViT-S/8	21	180	79.7	78.3
SCLRv2 [12]	RN152w3+SK	794	46	79.8	73.1
DINO	ViT-B/8	85	63	80.1	77.4

Source: Caron et al., “Emerging Properties in Self-Supervised Vision Transformers,” ICCV 2021, Table 2.

Supervised



DINO



	Random	Supervised	DINO
ViT-S/16	22.0	27.3	45.9
ViT-S/8	21.8	23.7	44.7

Figure 4: **Segmentations from supervised versus DINO.** We visualize masks obtained by thresholding the self-attention maps to keep 60% of the mass. On top, we show the resulting masks for a ViT-S/8 trained with supervision and DINO. We show the best head for both models. The table at the bottom compares the Jacard similarity between the ground truth and these masks on the validation images of PASCAL VOC12 dataset.

Source: Caron et al., “Emerging Properties in Self-Supervised Vision Transformers,” ICCV 2021, Fig. 4.

What makes DINO special?

- ▶ Learning from both big and small image parts helps the model understand local details.
- ▶ Using soft targets (from the teacher) gives the student more useful feedback than just using hard negatives.

Contrastive Learning: **DINOv2: Learning Robust Visual Features without Supervision**

DINOv2: Learning Robust Visual Features without Supervision

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Huy V. Vo*, Marc Szafraniec*, Vasil Khalidov*, Pierre Fernandez, Daniel Haziza,
Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba,
Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat,
Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal¹,
Patrick Labatut*, Armand Joulin*, Piotr Bojanowski*

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	INet-1k k-NN	INet-1k linear
iBOT	72.9	82.3
+(our reproduction)	74.5 $\uparrow 1.6$	83.2 $\uparrow 0.9$
+LayerScale, Stochastic Depth	75.4 $\uparrow 0.9$	82.0 $\downarrow 1.2$
+128k prototypes	76.6 $\uparrow 1.2$	81.9 $\downarrow 0.1$
+KoLeo	78.9 $\uparrow 2.3$	82.5 $\uparrow 0.6$
+SwiGLU FFN	78.7 $\downarrow 0.2$	83.1 $\uparrow 0.6$
+Patch size 14	78.9 $\uparrow 0.2$	83.5 $\uparrow 0.4$
+Teacher momentum 0.994	79.4 $\uparrow 0.5$	83.6 $\uparrow 0.1$
+Tweak warmup schedules	80.5 $\uparrow 1.1$	83.8 $\uparrow 0.2$
+Batch size 3k	81.7 $\uparrow 1.2$	84.7 $\uparrow 0.9$
+Sinkhorn-Knopp	81.7 =	84.7 =
+Untying heads = DINOv2	82.0 $\uparrow 0.3$	84.5 $\downarrow 0.2$

Source: Oquab et al., “DINOv2: Learning Robust Visual Features without Supervision,” 2023, Table 1.

Method	Arch.	Data	Text sup.	kNN	linear		
				val	val	ReaL	V2
Weakly supervised							
CLIP	ViT-L/14	WIT-400M	✓	79.8	84.3	88.1	75.3
CLIP	ViT-L/14 ₃₃₆	WIT-400M	✓	80.5	85.3	88.8	75.8
SWAG	ViT-H/14	IG3.6B	✓	82.6	85.7	88.7	77.6
OpenCLIP	ViT-H/14	LAION-2B	✓	81.7	84.4	88.4	75.5
OpenCLIP	ViT-G/14	LAION-2B	✓	83.2	86.2	89.4	77.2
EVA-CLIP	ViT-g/14	custom*	✓	83.5	86.4	89.3	77.4
Self-supervised							
MAE	ViT-H/14	INet-1k	✗	49.4	76.6	83.3	64.8
DINO	ViT-S/8	INet-1k	✗	78.6	79.2	85.5	68.2
SEERv2	RG10B	IG2B	✗	–	79.8	–	–
MSN	ViT-L/7	INet-1k	✗	79.2	80.7	86.0	69.7
EsViT	Swin-B/W=14	INet-1k	✗	79.4	81.3	87.0	70.4
Mugs	ViT-L/16	INet-1k	✗	80.2	82.1	86.9	70.8
iBOT	ViT-L/16	INet-22k	✗	72.9	82.3	87.5	72.4
DINOv2	ViT-S/14	LVD-142M	✗	79.0	81.1	86.6	70.9
	ViT-B/14	LVD-142M	✗	82.1	84.5	88.3	75.1
	ViT-L/14	LVD-142M	✗	83.5	86.3	89.5	78.0
	ViT-g/14	LVD-142M	✗	83.5	86.5	89.6	78.4

Source: Oquab et al., “DINOv2: Learning Robust Visual Features without Supervision,” 2023, Table 4.

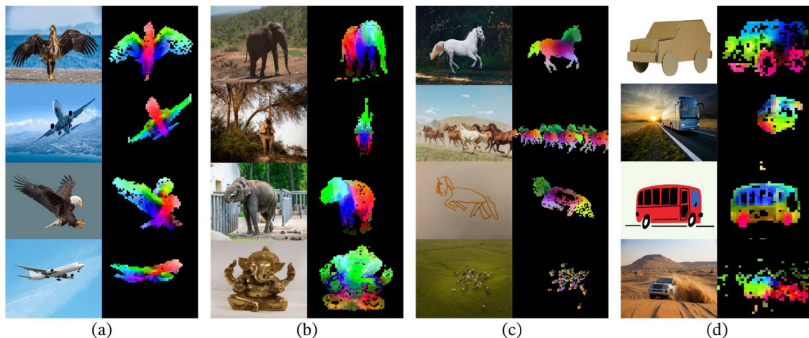


Figure 1: **Visualization of the first PCA components.** We compute a PCA between the patches of the images from the same column (a, b, c and d) and show their first 3 components. Each component is matched to a different color channel. Same parts are matched between related images despite changes of pose, style or even objects. Background is removed by thresholding the first PCA component.

Source: Oquab et al., “DINOv2: Learning Robust Visual Features without Supervision,” 2023, Fig. 1.

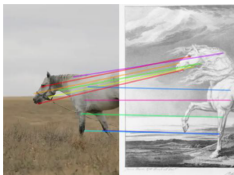


Figure 1: Feature matching

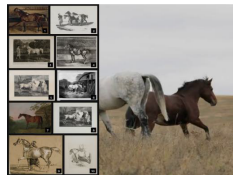


Figure 2: Image retrieval



Figure 3: Segmentation

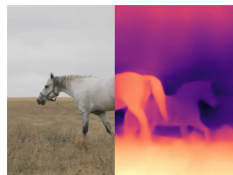


Figure 4: Depth prediction

Source: Meta AI, DINOv2 interactive demo, <https://dinov2.metademolab.com/>.

Contrastive Learning: **Image BERT Pre-Training with Online Tokenizer (iBOT)**

iBOT : IMAGE BERT PRE-TRAINING WITH ONLINE TOKENIZER

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Why iBOT?

- ▶ Improves self-supervised learning for vision transformers.
- ▶ Combines two main ideas:
 - Learning from missing pieces of images (like solving a puzzle).
 - Learning by copying a smart teacher model.

How does it work?

- ▶ **Masked Image Modeling (MIM):** Randomly hide some image patches, so the model has to guess what's missing.
- ▶ **Online Distillation:** There's a student model and a teacher model. The student tries to predict both the overall image and the hidden patches, using hints from the teacher.
- ▶ **Teacher as EMA:** The teacher isn't trained directly. Instead, it's a slow-moving average of the student, just like in DINO or BYOL.
- ▶ **Losses:**
 - **Image-level distillation:** The student learns to match the teacher's understanding of the whole image.
 - **Token-level distillation:** The student also learns to match the teacher's guesses for the hidden patches.

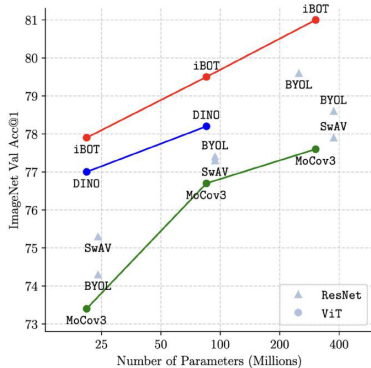


Table 6: **Object detection (Det.) & instance segmentation (ISeg.) on COCO and Semantic segmentation (Seg.) on ADE20K.** We report the results of ViT-S/16 (left) and ViT-B/16 (right). Seg.[†] denotes using a linear head for semantic segmentation.

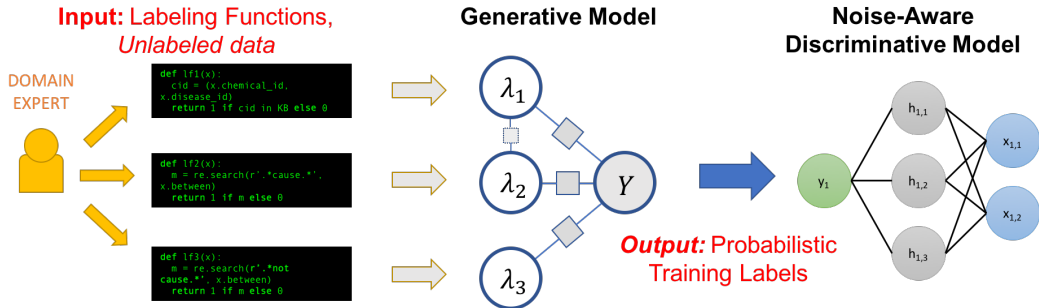
Method	Arch.	Param.	Det.	ISeg.	Seg.	Method	Det.	ISeg.	Seg. [†]	Seg.
			AP ^b	AP ^m	mIoU		AP ^b	AP ^m	mIoU	mIoU
Sup.	Swin-T	29	48.1	41.7	44.5	Sup.	49.8	43.2	35.4	46.6
MoBY	Swin-T	29	48.1	41.5	44.1	BEiT	50.1	43.5	27.4	45.8
Sup.	ViT-S/16	21	46.2	40.1	44.5	DINO	50.1	43.4	34.5	46.8
iBOT	ViT-S/16	21	49.4	42.6	45.4	iBOT	51.2	44.2	38.3	50.0

Source: Zhou et al., “iBOT: Image BERT Pre-Training with Online Tokenizer,” ICLR 2022, Fig. 1 (left) and Table 6 (right).

Why is this cool?

- ▶ Mixing MIM and distillation helps the model learn both big-picture (global) and detailed (local) features.
- ▶ iBOT gives strong linear-probe and fine-tuning results on ImageNet-1K (82.3% linear, 87.8% fine-tuned) and transfers well to dense tasks such as detection and segmentation.

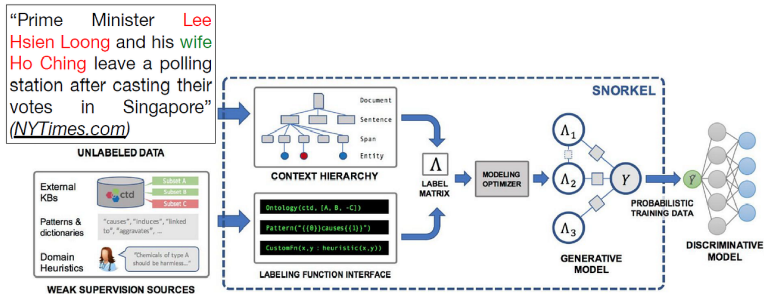
Weakly Supervised Learning (WSL)



Source: A. Ratner, P. Varma, B. Hancock, C. Ré et al., “Weak Supervision: A New Programming Paradigm for Machine Learning,” Stanford AI Lab blog, 2019.

- ▶ Sometimes, we don't have perfect data or labels for training.
- ▶ **Weakly Supervised Learning (WSL)** helps us learn even when our labels are not ideal.
- ▶ There are different ways labels can be “weak”:
 - **Incomplete labels:** Only some data points have labels.
 - **Inexact labels:** Labels are not very detailed (e.g., we know what's in an image, but not where).
 - **Inaccurate labels:** Some labels might be wrong or noisy.

Snorkel: data programming



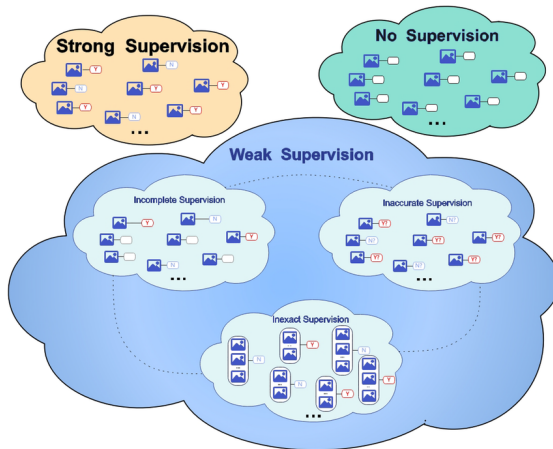
Source: K. Khvatova, "Weakly Supervised Learning: Introduction and Best Practices," Data Science Milan, 2019 (slides).

Why do we use WSL?

- ▶ Getting perfect labels for all data is expensive and slow.
- ▶ Experts are needed to label some data, which costs a lot.
- ▶ WSL lets us use cheaper, less perfect labels to train our models.

When is WSL useful?

- ▶ When we have lots of data but not enough labels (like in medical images).
- ▶ When we can use rules or existing knowledge to help label data (like using a dictionary for text).



Adapted from Zhi-Hua Zhou, "A Brief Introduction to Weakly Supervised Learning," *National Science Review* 5(1), 2018, Fig. 1.

How can we do Weakly Supervised Learning?

► Programmatic Labeling (Data Programming)

- Write simple rules (labeling functions) to label data automatically.
- Use a tool like Snorkel to combine these noisy labels and clean them up.
- Steps: Unlabeled data $\rightarrow$ Rules $\rightarrow$ Cleaned labels $\rightarrow$ Train model.
- (Snorkel: Ratner et al., 2017)

► Multiple Instance Learning (MIL)

- Sometimes we only know if a group (bag) has something, not which item.
- Example: We know an image has a cat, but not where.
- The model learns to find the right items inside each group.
- Used in medical images, object detection, etc.

► Webly / Noisy Label Learning

- Get labels from the internet or crowdsourcing; they can be messy!
- Use special tricks to handle wrong or noisy labels (like Co-teaching).

► Hybrid Methods (Semi + Weak Supervision)

- Mix a small set of good labels with lots of weak ones.
- Use tricks like pseudo-labeling or learning in steps.

► SimCLR:

- Simple framework for contrastive learning.
- Uses data augmentation and contrastive loss.

► BYOL:

- Self-supervised approach without negative pairs.
- Uses two networks: online and target.

► DINO:

- Self-distillation method without labels.
- Uses teacher-student networks and knowledge distillation.

► DINOv2:

- Improved version of DINO.
- Enhanced training strategies and architectures.
- Better performance and scalability.

► iBOT:

- Combines contrastive learning with masked image modeling.
- Uses BERT-style pretraining and vision transformers.

Contrastive Learning: **References**

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- [8] Zhi-Hua Zhou. *A Brief Introduction to Weakly Supervised Learning*. National Science Review, 5(1), 2018.
- [9] Alex Ratner et al. *Snorkel: Rapid Training Data Creation with Weak Supervision*. Proceedings of the VLDB Endowment, 2017. [arXiv:1711.10160](https://arxiv.org/abs/1711.10160).
- [10] Alex Ratner, Paroma Varma, Braden Hancock, Chris Ré et al. *Weak Supervision: A New Programming Paradigm for Machine Learning*. Stanford AI Lab blog, 2019. ai.stanford.edu/blog/weak-supervision.

- [11] **Related:** Xinlei Chen, Kaiming He. *Exploring Simple Siamese Representation Learning*. CVPR 2021. [arXiv:2011.10566](#).
- [12] **Tutorial:** Yann LeCun's Lectures on Self-Supervised Learning (YouTube).
- [13] **Code:** [SimCLR \(TensorFlow\)](#)
- [14] **Code:** [BYOL \(PyTorch\)](#)
- [15] **Code:** [DINO \(PyTorch\)](#)
- [16] **Code:** [iBOT \(PyTorch\)](#)